

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #4

A Gallery Walk of Industrial Engineering Problems – Cont'd

Ceyda Oğuz
coguz@ku.edu.tr

Koç University

13

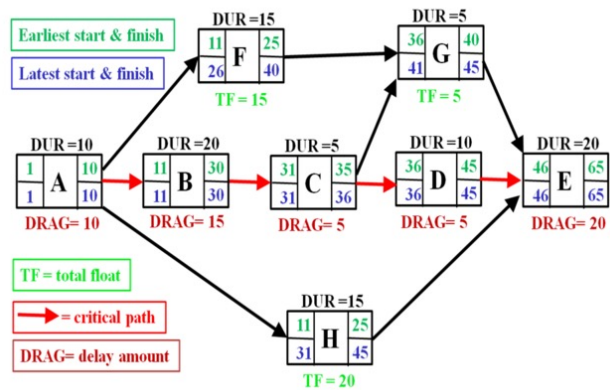
The Critical Path Problem

- The Impossible Build: Roof before walls?
- Precedence Constraints: Some tasks must wait.
- Parallel vs. Sequential tasks.

14

The Red Wire (Critical Path)

- The Critical Path (Red arrows) has zero slack.
- Delay here = Delay everywhere.
- The Job: Find the red wire and protect it.



This Photo by Unknown Author is licensed under CC BY-SA

15

The Critical Path Problem

- A country is preparing to host the **Summer Olympics**.
- **The deadline is fixed** — opening ceremony day cannot move.
- There are many tasks to complete
- Billions of viewers are watching, and a delay in a single task can become global news.

Olympic Project Timeline

Task	Description	Duration (months)	Depends on
	Build main Olympic stadium	30	—
	Construct athletes' village	24	—
	Develop transport infrastructure	20	A
	Build aquatic & sports venues	18	A
	Install IT, broadcast & security	10	C, D
	Test events & rehearsals	6	B, E
	Opening ceremony preparations	4	F

16

The Critical Path Problem

Data

- V : set of activities (tasks)
- $E \subseteq V \times V$: precedence arcs ($i \rightarrow j$) meaning activity j can start only after activity i finishes
- $d_i \geq 0$: duration of activity i

Decision variables

- $s_i \geq 0$: start time of activity i
- $T \geq 0$: project completion time (makespan)

Model

$$\begin{array}{ll}
 \text{minimize} & T \\
 \text{s.t.} & s_j \geq s_i + d_i \quad \forall (i, j) \in E \text{ (precedence)} \\
 & T \geq s_i + d_i \quad \forall i \in V \text{ (project ends after all)} \\
 & s_i \geq 0 \quad \forall i \in V
 \end{array}$$

17

Set Covering Problem

Motivation

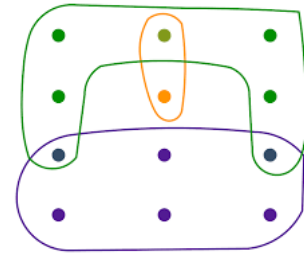
- Many real-life problems require choosing the *minimum number of resources* to cover all needs.
- Examples:
 - Placing the fewest number of Wi-Fi routers to cover a campus.
 - Selecting the minimum number of ambulance stations to cover a city.
 - Choosing minimal inspection points to monitor a pipeline network.

18

Set Covering Problem

Definition

- Given: A universe $U = \{1, \dots, n\}$ of elements to cover.
- A collection of sets $S_1, S_2, \dots, S_m$, each containing some elements of U .
- Each set S_i has a cost $c_i, i = 1, \dots, m$.
- Goal:
Find the smallest collection of sets whose union covers all elements of U .



19

Set Covering Problem

Mathematical Formulation

Define binary decision variable:

$$x_i = \begin{cases} 1, & \text{if collection } S_i, i = 1, \dots, m, \text{ is selected} \\ 0, & \text{otherwise.} \end{cases}$$

Objective:

$$\text{minimize } \sum_{i=1}^m c_i x_i$$

Subject to coverage constraints:

$$\sum_{i: j \in S_i} x_i \geq 1 \quad \forall j \in U$$

20

Set Covering Problem: Greedy Heuristic

- **Initialize:**
 - $C = \emptyset$ (the collection of chosen sets)
 - $U' = U$ (the set of uncovered elements)
- **Repeat until all elements are covered:**
 - For each remaining set S_j , compute:

$$\text{Efficiency}(S_j) = \frac{|\text{elements of } S_j \text{ not yet covered}|}{c_j}$$
 - Select the set S_j^* with the **highest efficiency**.
 - Add S_j^* to C .
 - Remove its elements from U' (mark them as covered).
- **Stop** when $U' = \emptyset$.

21

The Binary Switch

- We use Binary Variables (0 or 1).
- 1 = YES
- 0 = NO
- We turn the problem into a sequence of Yes/No decisions.

23